

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$y = x^3 - 10x^2 + 27x - 23 \Rightarrow \frac{dy}{dx} = 3x^2 - 20x + 27$	B1	1.1b
	$\left(\frac{dy}{dx}\right)_{x=5} = 3 \times 5^2 - 20 \times 5 + 27 (= 2)$	M1	1.1b
	$y + 13 = 2(x - 5)$	M1	2.1
	$y = 2x - 23$	A1	1.1b
	(4)		
(b)	Both C and l pass through $(0, -23)$ and so C meets l again on the y -axis	B1	2.2a
	(1)		

(c)	$\pm \int (x^3 - 10x^2 + 27x - 23 - (2x - 23)) \, dx$ $= \pm \left(\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{25}{2}x^2 \right)$	M1 A1ft	1.1b 1.1b
	$\left[\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{25}{2}x^2 \right]_0^5$ $= \left(\frac{625}{4} - \frac{1250}{3} + \frac{625}{2} \right) (-0)$	dM1	2.1
	$= \frac{625}{12}$	A1	1.1b
	(4)		
(c) Alternative 1:			
	$\pm \int (x^3 - 10x^2 + 27x - 23) \, dx$ $= \pm \left(\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 - 23x \right)$	M1 A1	1.1b 1.1b
	$\left[\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 - 23x \right]_0^5 + \frac{1}{2} \times 5(23 + 13)$ $= -\frac{455}{12} + 90$	dM1	2.1
	$= \frac{625}{12}$	A1	1.1b
	(4)		
(c) Alternative 2:			
	$\int (x^3 - 10x^2 + 27x) \, dx = \left(\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 \right)$	M1 A1	1.1b 1.1b
	$\left[\frac{x^4}{4} - \frac{10}{3}x^3 + \frac{27}{2}x^2 \right]_0^5 - \frac{1}{2} \times 5 \times 10$	dM1	2.1
	$= \frac{625}{12}$	A1	1.1b
	(4)		
(9 marks)			

(a)

B1: Correct derivative

M1: Substitutes $x = 5$ into their derivative. This may be implied by their value for $\frac{dy}{dx}$

M1: Fully correct straight line method using $(5, -13)$ and their $\frac{dy}{dx}$ at $x = 5$

A1: cao. Must see the full equation in the required form.

(b)

B1: Makes a suitable deduction.

Alternative via equating l and C and factorising e.g.

$$x^3 - 10x^2 + 27x - 23 = 2x - 23$$

$$x^3 - 10x^2 + 25x = 0$$

$$x(x^2 - 10x + 25) = 0 \Rightarrow x = 0$$

So they meet on the y -axis

(c)

M1: For an attempt to integrate $x^n \rightarrow x^{n+1}$ for $\pm C - l$

A1ft: Correct integration in any form which may be simplified or unsimplified. (follow through their equation from (a))

If they attempt as 2 separate integrals e.g. $\int (x^3 - 10x^2 + 27x - 23) dx - \int (2x - 23) dx$ then

award this mark for the correct integration of the curve as in the alternative.

If they combine the curve with the line first then the subsequent integration must be correct or a correct fit for their line and allow for $\pm C - l$

dm1: Fully correct strategy for the area. Award for use of 5 as the limit and condone the omission of the -0 . **Depends on the first method mark.**

A1: Correct exact value

Alternative 1:

M1: For an attempt to integrate $x^n \rightarrow x^{n+1}$ for $\pm C$

A1: Correct integration for $\pm C$

dm1: Fully correct strategy for the area e.g. correctly attempts the area of the trapezium and subtracts the area enclosed between the curve and the x -axis. Need to see the use of 5 as the limit condoning the omission of the -0 and a correct attempt at the trapezium and the subtraction.

May see the trapezium area attempted as $\int (2x - 23) dx$ in which case the integration and

use of the limits needs to be correct or correct follow through for their straight line equation.

Depends on the first method mark.

A1: Correct exact value

Note if they do $l - C$ rather than $C - l$ and the working is otherwise correct allow full marks if their final answer is given as a positive value. E.g. correct work with $l - C$ leading to $-\frac{625}{12}$ and

then e.g. hence area is $\frac{625}{12}$ is acceptable for full marks.

If the answer is left as $-\frac{625}{12}$ then score A0

Alternative 2:

M1: For an attempt to integrate $x^n \rightarrow x^{n+1}$ for $(C + 23)$

A1: Correct integration for $(C + 23)$

dM1: Fully correct strategy for the area e.g. correctly attempts the area of the triangle and subtracts from the area under the curve

Need to see the use of 5 as the limit condoning the omission of the “- 0” and a correct attempt at the triangle and the subtraction.

Depends on the first method mark.

Q2.

Question	Scheme	Marks	AOs
(a)(i)	$(x-5)^2 + (y+2)^2 = \dots$	M1	1.1b
	$(5, -2)$	A1	1.1b
(ii)	$r = \sqrt{5^2 + (-2)^2 - 11}$	M1	1.1b
	$r = 3\sqrt{2}$	A1	1.1b
		(4)	
(b)	$y = 3x + k \Rightarrow x^2 + (3x + k)^2 - 10x + 4(3x + k) + 11 = 0$ $\Rightarrow x^2 + 9x^2 + 6kx + k^2 - 10x + 12x + 4k + 11 = 0$	M1	2.1
	$\Rightarrow 10x^2 + (6k + 2)x + k^2 + 4k + 11 = 0$	A1	1.1b
	$b^2 - 4ac = 0 \Rightarrow (6k + 2)^2 - 4 \times 10 \times (k^2 + 4k + 11) = 0$	M1	3.1a
	$\Rightarrow 4k^2 + 136k + 436 = 0 \Rightarrow k = \dots$	M1	1.1b
	$k = -17 \pm 6\sqrt{5}$	A1	2.2a
		(5)	
(9 marks)			
Notes			

(a)(i)

M1: Attempts to complete the square on by halving both x and y terms.

Award for sight of $(x \pm 5)^2, (y \pm 2)^2 = \dots$ This mark can be implied by a centre of $(\pm 5, \pm 2)$.

A1: Correct coordinates. (Allow $x = 5, y = -2$)

(a)(ii)

M1: Correct strategy for the radius or radius². For example award for $r = \sqrt{\pm 5^2 + \pm 2^2 - 11}$

or an attempt such as $(x-a)^2 - a^2 + (y-b)^2 - b^2 + 11 = 0 \Rightarrow (x-a)^2 + (y-b)^2 = k \Rightarrow r^2 = k$

A1: $r = 3\sqrt{2}$. Do not accept for the A1 either $r = \pm 3\sqrt{2}$ or $\sqrt{18}$

The A1 can be awarded following sign slips on $(5, -2)$ so following $r^2 = \pm 5^2 + \pm 2^2 - 11$

(b) Main method seen

M1: Substitutes $y = 3x + k$ into the given equation (or their factorised version) and makes progress by attempting to expand the brackets. Condone lack of $= 0$

A1: Correct 3 term quadratic equation.

The terms must be collected but this can be implied by correct a, b and c

M1: Recognises the requirement to use $b^2 - 4ac = 0$ (or equivalent) where both b and c are expressions in k . It is dependent upon having attempted to substitute $y = 3x + k$ into the given equation

M1: Solves 3TQ in k . See General Principles.

The 3TQ in k must have been found as a result of attempt at $b^2 - 4ac \dots 0$

A1: Correct simplified values

Look carefully at the method used. It is possible to attempt this using gradients

(b) Alt 1	$x^2 + y^2 - 10x + 4y + 11 = 0 \Rightarrow 2x + 2y \frac{dy}{dx} - 10 + 4 \frac{dy}{dx} = 0$	M1 A1	2.1 1.1b
	Sets $\frac{dy}{dx} = 3 \Rightarrow x + 3y + 1 = 0$ and combines with equation for C $\Rightarrow 5x^2 - 50x + 44 = 0$ or $5y^2 + 20y + 11 = 0$ $\Rightarrow x = \dots$ or $y = \dots$	M1	3.1a
	$x = \frac{25 \pm 9\sqrt{5}}{5}, y = \frac{-10 \pm 3\sqrt{5}}{5}, k = y - 3x \Rightarrow k = \dots$	M1	1.1b
	$k = -17 \pm 6\sqrt{5}$	A1	2.2a

M1: Differentiates implicitly condoning slips but must have two $\frac{dy}{dx}$'s coming from correct terms

A1: Correct differentiation.

M1: Sets $\frac{dy}{dx} = 3$, makes y or x the subject, substitutes back into C and attempts to solve the resulting quadratic in x or y .

M1: Uses at least one pair of coordinates and l to find at least one value for k . It is dependent upon having attempted both M's

A1: Correct simplified values

(b) Alt 2	$x^2 + y^2 - 10x + 4y + 11 = 0 \Rightarrow 2x + 2y \frac{dy}{dx} - 10 + 4 \frac{dy}{dx} = 0$	M1 A1	2.1 1.1b
	Sets $\frac{dy}{dx} = 3 \Rightarrow x + 3y + 1 = 0$ and combines with equation for l $y = 3x + k, x + 3y = 1$ $\Rightarrow x = \dots$ and $y = \dots$ in terms of k	M1	3.1a
	$x = \frac{-3k-1}{10}, y = \frac{k-3}{10}, x^2 + y^2 - 10x + 4y + 11 = 0 \Rightarrow k = \dots$	M1	1.1b
	$k = -17 \pm 6\sqrt{5}$	A1	2.2a

Very similar except it uses equation for l instead of C in mark 3

M1 A1: Correct differentiation (See alt 1)

M1: Sets $\frac{dy}{dx} = 3$, makes y or x the subject, substitutes back into l to obtain x and y in terms of k

M1: Substitutes for x and y into C and solves resulting 3TQ in k

A1: Correct simplified values

(b) Alt 3	$y = 3x + k \Rightarrow m = 3 \Rightarrow m_r = -\frac{1}{3}$	M1
	$y + 2 = -\frac{1}{3}(x - 5)$	A1
	$(x - 5)^2 + (y + 2)^2 = 18, \quad y + 2 = -\frac{1}{3}(x - 5)$ $\Rightarrow \frac{10}{9}(x - 5)^2 = 18 \Rightarrow x = \dots$ or $\Rightarrow 10(y + 2)^2 = 18 \Rightarrow y = \dots$	M1
	$x = \frac{25 \pm 9\sqrt{5}}{5}, \quad y = \frac{-10 \pm 3\sqrt{5}}{5}, \quad k = y - 3x \Rightarrow k = \dots$	M1
	$k = -17 \pm 6\sqrt{5}$	A1

M1: Applies negative reciprocal rule to obtain gradient of radius

A1: Correct equation of radial line passing through the centre of C

M1: Solves simultaneously to find x or y

Alternatively solves " $y = -\frac{1}{3}x - \frac{1}{3}$ " and $y = 3x + k$ to get x in terms of k which they substitute in

$x^2 + (3x + k)^2 - 10x + 4(3x + k) + 11 = 0$ to form an equation in k .

M1: Applies $k = y - 3x$ with at least one pair of values to find k

A1: Correct simplified values

Q3.

Question	Scheme	Marks	AOs
(a)	(i) $\frac{dy}{dx} = 12x^3 - 24x^2$	M1 A1	1.1b 1.1b
	(ii) $\frac{d^2y}{dx^2} = 36x^2 - 48x$	A1ft	1.1b
		(3)	
(b)	Substitutes $x = 2$ into their $\frac{dy}{dx} = 12 \times 2^3 - 24 \times 2^2$	M1	1.1b
	Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point"	A1	2.1
		(2)	
(c)	Substitutes $x = 2$ into their $\frac{d^2y}{dx^2} = 36 \times 2^2 - 48 \times 2$	M1	1.1b
	$\frac{d^2y}{dx^2} = 48 > 0$ and states "hence the stationary point is a minimum"	A1ft	2.2a
		(2)	
(7 marks)			

Notes:

(a)(i)

M1: Differentiates to a cubic form

A1: $\frac{dy}{dx} = 12x^3 - 24x^2$

(a)(ii)

A1ft: Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx} = 36x^2 - 48x$

(b)

M1: Substitutes $x = 2$ into their $\frac{dy}{dx}$

A1: Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" All aspects of the proof must be correct

(c)

M1: Substitutes $x = 2$ into their $\frac{d^2y}{dx^2}$

Alternatively calculates the gradient of C either side of $x = 2$

A1ft: For a correct calculation, a valid reason and a correct conclusion.

Follow through on an incorrect $\frac{d^2y}{dx^2}$

Q4.

Question	Scheme	Marks	AOs
(a)	Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$	M1	1.1b
	$\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} \quad (= 2\sqrt{3} \cos t)$	A1	1.1b
	(2)		
(b)	Substitutes $t = \frac{2\pi}{3}$ in $\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} = (-\sqrt{3})$	M1	2.1
	Uses gradient of normal $= -\frac{1}{dy/dx} = \left(\frac{1}{\sqrt{3}}\right)$	M1	2.1
	Coordinates of $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$	B1	1.1b
	Correct form of normal $y + \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}(x+1)$	M1	2.1
	Completes proof $\Rightarrow 2x - 2\sqrt{3}y - 1 = 0$ *	A1*	1.1b
	(5)		
(c)	Substitutes $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$	M1	3.1a
	Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic in $\cos t$	M1	3.1a
	$\Rightarrow 12\cos^2 t - 4\cos t - 5 = 0$	A1	1.1b
	Finds $\cos t = \frac{5}{6}, \frac{1}{2}$	M1	2.4
	Substitutes their $\cos t = \frac{5}{6}$ into $x = 2\cos t$, $y = \sqrt{3} \cos 2t$,	M1	1.1b
	$Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$	A1	1.1b
	(6)		
(13 marks)			

Notes:

(a)

M1: Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ and achieves a form $k \frac{\sin 2t}{\sin t}$ Alternatively candidates may apply the double angle identity for $\cos 2t$ and achieve a form $k \frac{\sin t \cos t}{\sin t}$

A1: Scored for a correct answer, either $\frac{\sqrt{3} \sin 2t}{\sin t}$ or $2\sqrt{3} \cos t$

(b)

M1: For substituting $t = \frac{2\pi}{3}$ in their $\frac{dy}{dx}$ which must be in terms of t

M1: Uses the gradient of the normal is the negative reciprocal of the value of $\frac{dy}{dx}$. This may be seen in the equation of l .

B1: States or uses (in their tangent or normal) that $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$

M1: Uses their numerical value of $-1/\frac{dy}{dx}$ with their $\left(-1, -\frac{\sqrt{3}}{2}\right)$ to form an equation of the normal at P

A1*: This is a proof and all aspects need to be correct. Correct answer only $2x - 2\sqrt{3}y - 1 = 0$

(c)

M1: For substituting $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$ to produce an equation in t . Alternatively candidates could use $\cos 2t = 2\cos^2 t - 1$ to set up an equation of the form $y = Ax^2 + B$.

M1: Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic equation in $\cos t$
In the alternative method it is for combining their $y = Ax^2 + B$ with $2x - 2\sqrt{3}y - 1 = 0$ to get an equation in just one variable

A1: For the correct quadratic equation $12\cos^2 t - 4\cos t - 5 = 0$

Alternatively the equations in x and y are $3x^2 - 2x - 5 = 0$ $12\sqrt{3}y^2 + 4y - 7\sqrt{3} = 0$

M1: Solves the quadratic equation in $\cos t$ (or x or y) and rejects the value corresponding to P .

M1: Substitutes their $\cos t = \frac{5}{6}$ or their $t = \arccos\left(\frac{5}{6}\right)$ in $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$

If a value of x or y has been found it is for finding the other coordinate.

A1: $Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$. Allow $x = \frac{5}{3}, y = \frac{7}{18}\sqrt{3}$ but do not allow decimal equivalents.

Q5.

Question	Scheme	Marks	AOs
	$C: y = x \ln x$; l is a normal to C at $P(e, e)$ Let x_A be the x -coordinate of where l cuts the x -axis		
	$\frac{dy}{dx} = \ln x + x \left(\frac{1}{x} \right) \quad \{= 1 + \ln x\}$	M1	2.1
		A1	1.1b
	$x = e, m_T = 2 \Rightarrow m_N = -\frac{1}{2} \Rightarrow y - e = -\frac{1}{2}(x - e)$ $y = 0 \Rightarrow -e = -\frac{1}{2}(x - e) \Rightarrow x = \dots$	M1	3.1a
	l meets x -axis at $x = 3e$ (allow $x = 2e + e \ln e$)	A1	1.1b
	{Areas:} either $\int_1^e x \ln x \, dx = [\dots]_1^e = \dots$ or $\frac{1}{2}((\text{their } x_A) - e)e$	M1	2.1
	$\left\{ \int x \ln x \, dx = \right\} \frac{1}{2} x^2 \ln x - \int \frac{1}{x} \left(\frac{x^2}{2} \right) \{dx\}$	M1	2.1
	$\left\{ = \frac{1}{2} x^2 \ln x - \int \frac{1}{2} x \{dx\} \right\} = \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2$	dM1	1.1b
		A1	1.1b
	$\text{Area}(R_1) = \int_1^e x \ln x \, dx = [\dots]_1^e = \dots$; $\text{Area}(R_2) = \frac{1}{2}((\text{their } x_A) - e)e$ and so, $\text{Area}(R) = \text{Area}(R_1) + \text{Area}(R_2) \quad \{= \frac{1}{4}e^2 + \frac{1}{4} + e^2\}$	M1	3.1a
	$\text{Area}(R) = \frac{5}{4}e^2 + \frac{1}{4}$	A1	1.1b
		(10)	

Notes for Question	
M1:	Differentiates by using the product rule to give $\ln x + x(\text{their } g'(x))$, where $g(x) = \ln x$
A1:	Correct differentiation of $y = x \ln x$, which can be un-simplified or simplified
M1:	Complete strategy to find the x coordinate where their normal to C at $P(e, e)$ meets the x -axis i.e. Sets $y=0$ in $y-e = m_N(x-e)$ to find $x = \dots$
Note:	m_T is found by using calculus and $m_N \neq m_T$
A1:	l meets x -axis at $x = 3e$, allowing un-simplified values for x such as $x = 2e + e \ln e$
Note:	Allow $x = \text{awrt } 8.15$
M1:	Scored for either - Area under curve $= \int_1^e x \ln x \, dx = [\dots]_1^e = \dots$, with limits of e and 1 and some attempt to substitute these and subtract - or Area under line $= \frac{1}{2}((\text{their } x_A) - e)e$, with a valid attempt to find x_A
M1:	Integration by parts the correct way around to give $Ax^2 \ln x - \int B \left(\frac{x^2}{x} \right) \{dx\}$; $A \neq 0, B > 0$
dM1:	dependent on the previous M mark Integrates the second term to give $\pm \lambda x^2$; $\lambda \neq 0$
A1:	$\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2$
M1:	Complete strategy of finding the area of R by finding the sum of two key areas. See scheme.
A1:	$\frac{5}{4}e^2 + \frac{1}{4}$
Note:	Area(R_2) can also be found by integrating the line l between limits of e and their x_A i.e. $\text{Area}(R_2) = \int_e^{\text{their } x_A} \left(-\frac{1}{2}x + \frac{3}{2}e \right) dx = [\dots]_e^{\text{their } x_A} = \dots$
Note:	Calculator approach with no algebra, differentiation or integration seen: - Finding l cuts through the x-axis at awrt 8.15 is 2nd M1 2nd A1 - Finding area between curve and the x-axis between $x=1$ and $x=e$ to give awrt 2.10 is 3rd M1 - Using the above information (must be seen) to apply $\text{Area}(R) = 2.0972\dots + 7.3890\dots = 9.4862\dots$ is final M1 Therefore, a maximum of 4 marks out of the 10 available.

Q6.

Question	Scheme	Marks	AOs
	$\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv A + \frac{B}{(x-3)} + \frac{C}{(1-2x)}$		
(a) Way 1	$1+11x-6x^2 \equiv A(1-2x)(x-3) + B(1-2x) + C(x-3) \Rightarrow B = \dots, C = \dots$	M1	2.1
	$A = 3$	B1	1.1b
	Uses substitution or compares terms to find either $B = \dots$ or $C = \dots$	M1	1.1b
	$B = 4$ and $C = -2$ which have been found using a correct identity	A1	1.1b
	(4)		
(a) Way 2	{long division gives} $\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv 3 + \frac{-10x+10}{(x-3)(1-2x)}$		
	$-10x+10 \equiv B(1-2x) + C(x-3) \Rightarrow B = \dots, C = \dots$	M1	2.1
	$A = 3$	B1	1.1b
	Uses substitution or compares terms to find either $B = \dots$ or $C = \dots$	M1	1.1b
	$B = 4$ and $C = -2$ which have been found using $-10x+10 \equiv B(1-2x) + C(x-3)$	A1	1.1b
	(4)		
(b)	$f(x) = 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}$ { $= 3 + 4(x-3)^{-1} - 2(1-2x)^{-1}$ } ; $x > 3$		
	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2} \left\{ = -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2} \right\}$	M1 A1ft	2.1 1.1b
	Correct $f'(x)$ and as $(x-3)^2 > 0$ and $(1-2x)^2 > 0$, then $f'(x) = -(+ve) - (+ve) < 0$, so $f(x)$ is a decreasing function	A1	2.4
	(3)		
(7 marks)			

Notes for Question	
(a)	
M1:	Way 1: Uses a correct identity $1+11x-6x^2 \equiv A(1-2x)(x-3) + B(1-2x) + C(x-3)$ in a complete method to find values for B and C . Note: Allow one slip in copying $1+11x-6x^2$ Way 2: Uses a correct identity $-10x+10 \equiv B(1-2x) + C(x-3)$ (which has been found from long division) in a complete method to find values for B and C
B1:	$A = 3$
M1:	Attempts to find the value of either B or C from their identity This can be achieved by <i>either</i> substituting values into their identity <i>or</i> by comparing coefficients and solving the resulting equations simultaneously
A1:	See scheme
Note:	Way 1: Comparing terms: $x^2: -6 = -2A$; $x: 11 = 7A - 2B + C$; constant: $1 = -3A + B - 3C$ Way 1: Substituting: $x = 3: -20 = -5B \Rightarrow B = 4$; $x = \frac{1}{2}: 5 = -\frac{5}{2}C \Rightarrow C = -2$
Note:	Way 2: Comparing terms: $x: -10 = -2B + C$; constant: $10 = B - 3C$ Way 2: Substituting: $x = 3: -20 = -5B \Rightarrow B = 4$; $x = \frac{1}{2}: 5 = -\frac{5}{2}C \Rightarrow C = -2$
Note:	$A = 3, B = 4, C = -2$ from no working scores M1B1M1A1
Note:	The final A1 mark is effectively dependent upon both M marks

Notes for Question Continued	
(a) ctd	
Note:	Writing $1+11x-6x^2 \equiv B(1-2x)+C(x-3) \Rightarrow B=4, C=-2$ will get 1 st M0, 2 nd M1, 1 st A0
Note:	Way 1: You can imply a correct identity $1+11x-6x^2 \equiv A(1-2x)(x-3)+B(1-2x)+C(x-3)$ from seeing $\frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{A(1-2x)(x-3)+B(1-2x)+C(x-3)}{(x-3)(1-2x)}$
Note:	Way 2: You can imply a correct identity $-10x+10 \equiv B(1-2x)+C(x-3)$ from seeing $\frac{-10x+10}{(x-3)(1-2x)} \equiv \frac{B(1-2x)+C(x-3)}{(x-3)(1-2x)}$
(b)	
M1:	Differentiates to give $\{f'(x) = \} \pm \lambda(x-3)^{-2} \pm \mu(1-2x)^{-2}; \lambda, \mu \neq 0$
A1ft:	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$, which can be simplified or un-simplified
Note:	Allow A1ft for $f'(x) = -(\text{their } B)(x-3)^{-2} + (2)(\text{their } C)(1-2x)^{-2}; (\text{their } B), (\text{their } C) \neq 0$
A1:	$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$ or $f'(x) = -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2}$ and a correct explanation e.g. $f'(x) = -(+ve) - (+ve) < 0$, so $f(x)$ is a decreasing {function}
Note:	The final A mark can be scored in part (b) from an incorrect $A = \dots$ or from $A = 0$ or no value of A found in part (a)

Notes for Question Continued - Alternatives					
(a)					
Note:	Be aware of the following alternative solutions, by initially dividing by " $(x-3)$ " or " $(1-2x)$ " $\bullet \frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{-6x-7}{(1-2x)} - \frac{20}{(x-3)(1-2x)} \equiv 3 - \frac{10}{(1-2x)} - \frac{20}{(x-3)(1-2x)}$ $\frac{20}{(x-3)(1-2x)} \equiv \frac{D}{(x-3)} + \frac{E}{(1-2x)} \Rightarrow 20 \equiv D(1-2x) + E(x-3) \Rightarrow D = -4, E = -8$ $\Rightarrow 3 - \frac{10}{(1-2x)} - \left(\frac{-4}{(x-3)} + \frac{-8}{(1-2x)} \right) \equiv 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}; A=3, B=4, C=-2$ $\bullet \frac{1+11x-6x^2}{(x-3)(1-2x)} \equiv \frac{3x-4}{(x-3)} + \frac{5}{(x-3)(1-2x)} \equiv 3 + \frac{5}{(x-3)} + \frac{5}{(x-3)(1-2x)}$ $\frac{5}{(x-3)(1-2x)} \equiv \frac{D}{(x-3)} + \frac{E}{(1-2x)} \Rightarrow 5 \equiv D(1-2x) + E(x-3) \Rightarrow D = -1, E = -2$ $\Rightarrow 3 + \frac{5}{(x-3)} + \left(\frac{-1}{(x-3)} + \frac{-2}{(1-2x)} \right) \equiv 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)}; A=3, B=4, C=-2$				
(b)					
Alternative Method 1:					
$f(x) = \frac{1+11x-6x^2}{(x-3)(1-2x)}, x > 3 \Rightarrow f(x) = \frac{1+11x-6x^2}{-2x^2+7x-3}; \begin{cases} u = 1+11x-6x^2 & v = -2x^2+7x-3 \\ u' = 11-12x & v' = -4x+7 \end{cases}$					
$f'(x) = \frac{(-2x^2+7x-3)(11-12x) - (1+11x-6x^2)(-4x+7)}{(-2x^2+7x-3)^2}$	<table border="1"> <tr> <td>Uses quotient rule to find $f'(x)$</td><td>M1</td></tr> <tr> <td>Correct differentiation</td><td>A1</td></tr> </table>	Uses quotient rule to find $f'(x)$	M1	Correct differentiation	A1
Uses quotient rule to find $f'(x)$	M1				
Correct differentiation	A1				
$f'(x) = \frac{-20((x-1)^2+1)}{(-2x^2+7x-3)^2}$ and a correct explanation, e.g. $f'(x) = -\frac{(+ve)}{(+ve)} < 0$, so $f(x)$ is a decreasing {function}	A1				
Alternative Method 2:					
Allow M1A1A1 for the following solution: Given $f(x) = 3 + \frac{4}{(x-3)} - \frac{2}{(1-2x)} = 3 + \frac{4}{(x-3)} + \frac{2}{(2x-1)}$ as $\frac{4}{(x-3)}$ decreases when $x > 3$ and $\frac{2}{(2x-1)}$ decreases when $x > 3$ then $f(x)$ is a decreasing {function}					

Q7.

Question	Scheme	Marks	AOs
(a)	$k = e^2$ or $x \neq e^2$	B1	2.2a
		(1)	
(b)	$g'(x) = \frac{(\ln x - 2) \times \frac{3}{x} - (3 \ln x - 7) \times \frac{1}{x}}{(\ln x - 2)^2} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = \frac{d}{dx} \left(3 - (\ln(x) - 2)^{-1} \right) = (\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = (\ln x - 2)^{-1} \times \frac{3}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$	M1 A1	1.1b 2.1
	As $x > 0$ (or $1/x > 0$) AND $\ln x - 2$ is squared so $g'(x) > 0$	A1cso	2.4
		(3)	
(c)	Attempts to solve either $3 \ln x - 7 \dots 0$ or $\ln x - 2 \dots 0$ or $3 \ln a - 7 \dots 0$ or $\ln a - 2 \dots 0$ where $\dots$ is “=” or “>” to reach a value for x or a but may be seen as an inequality e.g. $x > \dots$ or $a > \dots$	M1	3.1a
	$0 < a < e^2$, $a > e^{\frac{7}{3}}$	A1	2.2a
		(2)	
(6 marks)			

Notes:

(a)

B1: Deduces $k = e^2$ or $x \neq e^2$ Condone $k = \text{awrt } 7.39$ or $x \neq \text{awrt } 7.39$

(b)

M1: Attempts to differentiate via the quotient rule and with $\ln x \rightarrow \frac{1}{x}$ so allow for:

$$\frac{d}{dx}(g(x)) = \frac{(\ln x - 2) \times \frac{\alpha}{x} - (3 \ln x - 7) \times \frac{\beta}{x}}{(\ln x - 2)^2}, \quad \beta > 0$$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.

Alternatively attempts to write $g(x) = \frac{3 \ln(x) - 7}{\ln(x) - 2} = 3 - (\ln(x) - 2)^{-1}$ and attempts the chain rule so allow for:

$$3 - (\ln(x) - 2)^{-1} \rightarrow (\ln(x) - 2)^{-2} \times \frac{\alpha}{x}$$

Alternatively writes $g(x) = (3 \ln(x) - 7)(\ln(x) - 2)^{-1}$ and attempts the product rule so allow for:

$$g'(x) = (\ln x - 2)^{-1} \times \frac{\alpha}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{\beta}{x}$$

In general condone missing brackets for the M mark. E.g. if they quote $u = 3 \ln x - 7$ and $v = \ln x - 2$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: $\frac{1}{x(\ln x - 2)^2}$ Allow $\frac{\frac{1}{x}}{(\ln x - 2)^2}$ i.e. we need to see the numerator simplified to $1/x$

Note that some candidates establish the correct numerator and correct denominator independently and provided they obtain the correct expressions, this mark can be awarded.

But allow a correctly expanded denominator.

A1cso: States that as $x > 0$ AND $\ln x - 2$ is squared so $g'(x) > 0$



(c)

M1: Attempts to solve either $3\ln x - 7 = 0$ or $\ln x - 2 = 0$ or using inequalities e.g. $3\ln x - 7 > 0$

A1: $0 < a < e^2$, $a > e^{\frac{7}{3}}$

Q8.

Question	Scheme	Marks	AOs
(a)	$\frac{d}{dx}(3y^2) = 6y \frac{dy}{dx}$ or $\frac{d}{dx}(qxy) = qx \frac{dy}{dx} + qy$	M1	2.1
	$3px^2 + qx \frac{dy}{dx} + qy + 6y \frac{dy}{dx} = 0$	A1	1.1b
	$(qx + 6y) \frac{dy}{dx} = -3px^2 - qy \Rightarrow \frac{dy}{dx} = \dots$	dM1	2.1
	$\frac{dy}{dx} = \frac{-3px^2 - qy}{qx + 6y}$	A1	1.1b
	(4)		
(b)	$p(-1)^3 + q(-1)(-4) + 3(-4)^2 = 26$	M1	1.1b
	$19x + 26y + 123 = 0 \Rightarrow m = -\frac{19}{26}$	B1	2.2a
	$\frac{-3p(-1)^2 - q(-4)}{q(-1) + 6(-4)} = \frac{26}{19} \quad \text{or} \quad \frac{q(-1) + 6(-4)}{3p(-1)^2 + q(-4)} = -\frac{19}{26}$	M1	3.1a
	$p - 4q = 22, \quad 57p - 102q = 624 \Rightarrow p = \dots, q = \dots$	dM1	1.1b
	$p = 2, \quad q = -5$	A1	1.1b
	(5)		
(9 marks)			

Notes

(a)

M1: For selecting the appropriate method of differentiating:

Allow this mark for either $3y^2 \rightarrow \alpha y \frac{dy}{dx}$ or $qxy \rightarrow \alpha x \frac{dy}{dx} + \beta y$

A1: Fully correct differentiation. Ignore any spurious $\frac{dy}{dx} = \dots$

dM1: A valid attempt to make $\frac{dy}{dx}$ the subject with 2 terms only in $\frac{dy}{dx}$ coming from qxy and $3y^2$

Depends on the first method mark.

A1: Fully correct expression

(b)

M1: Uses $x = -1$ and $y = -4$ in the equation of C to obtain an equation in p and q

B1: Deduces the correct gradient of the given normal.

This may be implied by e.g.

$$19x + 26y + 123 = 0 \Rightarrow y = -\frac{19}{26}x + \dots \Rightarrow \text{Tangent equation is } y = \frac{26}{19}x + \dots$$

M1: Fully correct strategy to establish an equation connecting p and q using $x = -1$ and $y = -4$ in their $\frac{dy}{dx}$ and the gradient of the normal. E.g. $(a) = -1 \div \text{their } -\frac{19}{26}$ or $-1 \div (a) = \text{their } -\frac{19}{26}$

dM1: Solves simultaneously to obtain values for p and q .

Depends on both previous method marks.

A1: Correct values

Note that in (b), attempts to form the equation of the normal in terms of p and q and then compare coefficients with $19x + 26y + 123 = 0$ score no marks. If there is any doubt use Review.

Q9.

Question	Scheme	Marks	AOs
(a)	$\ln x \rightarrow \frac{1}{x}$	B1	1.1a
	Method to differentiate $\frac{4x^2 + x}{2\sqrt{x}}$ – see notes	M1	1.1b
	E.g. $2 \times \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$	A1	1.1b
	$\frac{dy}{dx} = 3\sqrt{x} + \frac{1}{4\sqrt{x}} - \frac{4}{x} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}} *$	A1*	2.1
	(4)		
(b)	$12x^2 + x - 16\sqrt{x} = 0 \Rightarrow 12x^{\frac{3}{2}} + x^{\frac{1}{2}} - 16 = 0$	M1	1.1b
	E.g. $12x^{\frac{3}{2}} = 16 - \sqrt{x}$	dM1	1.1b
	$x^{\frac{3}{2}} = \frac{4}{3} - \frac{\sqrt{x}}{12} \Rightarrow x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12}\right)^{\frac{2}{3}} *$	A1*	2.1
	(3)		
(c)	$x_2 = \sqrt[3]{\left(\frac{4}{3} - \frac{\sqrt{2}}{12}\right)^2}$	M1	1.1b
	$x_2 = \text{awrt } 1.13894$	A1	1.1b
	$x = 1.15650$	A1	2.2a
	(3)		
(10 marks)			

Notes:

(a)

B1: Differentiates $\ln x \rightarrow \frac{1}{x}$ seen or implied

M1: Correct method to differentiate $\frac{4x^2 + x}{2\sqrt{x}}$:

Look for $\frac{4x^2 + x}{2\sqrt{x}} \rightarrow \dots x^{\frac{3}{2}} + \dots x^{\frac{1}{2}}$ being then differentiated to $Px^{\frac{1}{2}} + \dots$ or $\dots + Qx^{-\frac{1}{2}}$

Alternatively uses the quotient rule on $\frac{4x^2 + x}{2\sqrt{x}}$.

Condone slips but if rule is not quoted expect $\left(\frac{dy}{dx}\right) = \frac{2\sqrt{x}(Ax+B) - (4x^2+x)Cx^{\frac{1}{2}}}{(2\sqrt{x})^2} (A, B, C > 0)$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.

Alternatively uses the product rule on $(4x^2 + x)(2\sqrt{x})^{-1}$

Condone slips but expect $\left(\frac{dy}{dx}\right) = Ax^{-\frac{1}{2}}(Bx+C) + D(4x^2+x)x^{-\frac{3}{2}} (A, B, C > 0)$

In general condone missing brackets for the M mark. If they quote $u = 4x^2 + x$ and $v = 2\sqrt{x}$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: Correct differentiation of $\frac{4x^2 + x}{2\sqrt{x}}$ although may not be simplified.

Examples: $\left(\frac{dy}{dx}\right) = \frac{2\sqrt{x}(8x+1) - (4x^2+x)x^{-\frac{1}{2}}}{(2\sqrt{x})^2}, \frac{1}{2}x^{-\frac{1}{2}}(8x+1) - \frac{1}{4}(4x^2+x)x^{-\frac{3}{2}}, 2 \times \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$

A1*: Obtains $\frac{dy}{dx} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}}$ via $3\sqrt{x} + \frac{1}{4\sqrt{x}} - \frac{4}{x}$ or a correct application of the quotient or product rule

and with sufficient working shown to reach the printed answer.

There must be no errors e.g. missing brackets.

(b)

M1: Sets $12x^2 + x - 16\sqrt{x} = 0$ and divides by $\sqrt{x}$ or equivalent e.g. divides by x and multiplies by $\sqrt{x}$

dM1: Makes the term in $x^{\frac{3}{2}}$ the subject of the formula

A1*: A correct and rigorous argument leading to the given solution.

Alternative - working backwards:

$$x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12}\right)^{\frac{2}{3}} \Rightarrow x^{\frac{3}{2}} = \frac{4}{3} - \frac{\sqrt{x}}{12} \Rightarrow 12x^{\frac{3}{2}} = 16 - \sqrt{x} \Rightarrow 12x^2 = 16\sqrt{x} - x \Rightarrow 12x^2 - 16\sqrt{x} + x = 0$$

M1: For raising to power of $3/2$ both sides. **dM1:** Multiplies through by $\sqrt{x}$. **A1:** Achieves printed answer and makes a minimal comment e.g. tick, #, QED, true etc.

(c)

M1: Attempts to use the iterative formula with $x_1 = 2$. This is implied by sight of $x_2 = \left(\frac{4}{3} - \frac{\sqrt{2}}{12}\right)^{\frac{2}{3}}$ or awrt 1.14

A1: $x_2 = \text{awrt } 1.13894$

A1: Deduces that $x = 1.15650$

Q10.

Question	Scheme	Marks	AOs
(a)	$y = \operatorname{cosec}^3 \theta \Rightarrow \frac{dy}{d\theta} = -3 \operatorname{cosec}^2 \theta \operatorname{cosec} \theta \cot \theta$	B1	1.1b
	$\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$	M1	1.1b
	$\frac{dy}{dx} = \frac{-3 \operatorname{cosec}^3 \theta \cot \theta}{2 \cos 2\theta}$	A1	1.1b
		(3)	
(b)	$y = 8 \Rightarrow \operatorname{cosec}^3 \theta = 8 \Rightarrow \sin^3 \theta = \frac{1}{8} \Rightarrow \sin \theta = \frac{1}{2}$	M1	3.1a
	$\theta = \frac{\pi}{6} \Rightarrow \frac{dy}{dx} = \frac{-3 \operatorname{cosec}^3 \left(\frac{\pi}{6} \right) \cot \left(\frac{\pi}{6} \right)}{2 \cos \left(\frac{2\pi}{6} \right)} = \dots$ or $\sin \theta = \frac{1}{2} \Rightarrow \frac{dy}{dx} = \frac{\frac{-3}{\sin^3 \theta} \times \frac{\cos \theta}{\sin \theta}}{2(1 - 2 \sin^2 \theta)} = \frac{-3 \times 8 \times \frac{\sqrt{3}/2}{1/2}}{2 \left(1 - 2 \times \frac{1}{4} \right)}$	M1	2.1
	$= -24\sqrt{3}$	A1	2.2a
		(3)	
(6 marks)			

Notes
(a) B1: Correct expression for $\frac{dy}{d\theta}$ seen or implied in any form e.g. $\frac{-3 \cos \theta}{\sin^4 \theta}$ M1: Obtains $\frac{dx}{d\theta} = k \cos 2\theta$ or $\alpha \cos^2 \theta + \beta \sin^2 \theta$ (from product rule on $\sin \theta \cos \theta$) and attempts $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ A1: Correct expression in any form. May see e.g. $\frac{-3 \cos \theta}{2 \sin^4 \theta \cos 2\theta}, -\frac{3}{4 \sin^4 \theta \cos \theta - 2 \sin^3 \theta \tan \theta}$ (b) M1: Recognises the need to find the value of $\sin \theta$ or θ when $y = 8$ and uses the y parameter to establish its value. This should be correct work leading to $\sin \theta = \frac{1}{2}$ or e.g. $\theta = \frac{\pi}{6}$ or 30°. M1: Uses their value of $\sin \theta$ or θ in their $\frac{dy}{dx}$ from part (a) (working in exact form) in an attempt to obtain an exact value for $\frac{dy}{dx}$. May be implied by a correct exact answer. If no working is shown but an exact answer is given you may need to check that this follows their $\frac{dy}{dx}$. A1: Deduces the correct gradient

Q11.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t$ or $5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2
	$(\text{Area}) = \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area}) = \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt$ *	A1*	2.1*
	(3)		
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = [-20 \cos^3 t]_0^{\frac{\pi}{2}} = 0 - (-20) = 20$ *	A1*	2.1
	(3)		
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986..., 1.072...$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986... \Rightarrow x = 2.869...$ $t = 1.072 \Rightarrow x = 5.269...$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
	(5)		
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

e.g. $A \int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “ dt ”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$$\int 60 \sin t \cos^2 t \, dt = k u^3$$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

not just: $-20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = ...$

A1: At least one correct value for t , correct to 2 dp. FYI $t = 0.4986..., 1.072...$ or in degrees $t = 28.57..., 61.42...$

dM1: Attempts to find **TWO** distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2

values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form A1: Correct quartic M1: Solves quartic A1: Correct values

A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. Units are required.

Q12.

Question	Scheme	Marks	AOs
(a)	$f(x) = 4(x^2 - 2)e^{-2x}$		
	Differentiates to $e^{-2x} \times 8x + 4(x^2 - 2) \times -2e^{-2x}$	M1 A1	1.1b 1.1b
	$f'(x) = 8e^{-2x} \{x - (x^2 - 2)\} = 8(2 + x - x^2)e^{-2x} *$	A1*	2.1
		(3)	
(b)	States roots of $f'(x) = 0$ $x = -1, 2$	B1	1.1b
	Substitutes one x value to find a y value	M1	1.1b
	Stationary points are $(-1, -4e^2)$ and $(2, 8e^{-4})$	A1	1.1b
		(3)	
(c)	(i) Range $[-8e^2, \infty)$ o.e. such as $g(x) \geq -8e^2$	B1ft	2.5
	(ii) For - Either attempting to find $2f(0) - 3 = 2 \times -8 - 3 = (-19)$ and identifying this as the lower bound - Or attempting to find $2 \times "8e^{-4}" - 3$ and identifying this as the upper bound	M1	3.1a
	Range $[-19, 16e^{-4} - 3]$	A1	1.1b
		(3)	
(9 marks)			
Notes:			

(a)

M1: Attempts the product rule and uses $e^{-2x} \rightarrow ke^{-2x}$, $k \neq 0$

If candidate states $u = 4(x^2 - 2)$, $v = e^{-2x}$ with $u' = \dots$, $v' = \dots e^{-2x}$ it can be implied by their $vu' + uv'$

If they just write down an answer without working award for $f'(x) = pxe^{-2x} \pm q(x^2 - 2)e^{-2x}$

They may multiply out first $f(x) = 4x^2e^{-2x} - 8e^{-2x}$. Apply in the same way condoning slips

Alternatively attempts the quotient rule on $f(x) = \frac{u}{v} = \frac{4(x^2 - 2)}{e^{2x}}$ with $v' = ke^{2x}$ and $f'(x) = \frac{vu' - uv'}{v^2}$

A1: A correct $f'(x)$ which may be unsimplified.

Via the quotient rule you can award for $f'(x) = \frac{8xe^{2x} - 8(x^2 - 2)e^{2x}}{e^{4x}}$ o.e.

A1*: Proceeds correctly to given answer showing all necessary steps.

The $f'(x)$ or $\frac{dy}{dx}$ must be present at some point in the solution

This is a "show that" question and there must not be any errors. All bracketing must be correct.

Allow a candidate to move from the **simplified** unfactorised answer of $f'(x) = 8xe^{-2x} - 8(x^2 - 2)e^{-2x}$

to the given answer in one step.

Do not allow it from an **unsimplified** $f'(x) = 4 \times 2xe^{-2x} + 4(x^2 - 2) \times -2e^{-2x}$

Allow the expression / bracketed expression to be written in a different order.

So, for example, $8(x - x^2 + 2)e^{-2x}$ is OK

(b)

B1: States or implies $x = -1, 2$ (as the roots of $f'(x) = 0$)

M1: Substitutes one x value of their solution to $f'(x) = 0$ in $f(x)$ to find a y value.

Allow decimals here (3sf). FYI, to 3 sf, $-4e^2 = -29.6$ and $8e^{-4} = 0.147$

Some candidates just write down the x coordinates but then go on in part (c) to find the ranges using the y coordinates. Allow this mark to be scored from work in part (c)

A1: Obtains $(-1, -4e^2)$ and $(2, 8e^{-4})$ as the stationary points. This must be scored in (b). Remember to isw

after a correct answer. Allow these to be written separately. E.g. $x = -1$, $y = -4e^2$

Extra solutions, e.g. from $x = 0$ will be penalised on this mark.

(c)(i)

B1ft: For a correct range written using correct notation.

Follow through on $2 \times$ their minimum "y" value from part (b), providing it is negative.

Condone a decimal answer if this is consistent with their answer in (b) to 3sf or better.

Examples of correct responses are $[-8e^2, \infty)$, $y \geq -8e^2$, $y \geq -8e^2$, $\{q \in \mathbb{R}, q \geq -8e^2\}$

(c)(ii)

M1: See main scheme. Follow through on $2 \times$ their " $8e^{-4}$ " - 3 for the upper bound.

A1: Range $[-19, 16e^{-4} - 3]$ o.e. such as $-19 \leq y \leq 16e^{-4} - 3$ but must be exact

Q13.

Question	Scheme	Marks	AOs
(a)	$x = u^2 + 1 \Rightarrow dx = 2u du$ oe	B1	1.1b
	Full substitution $\int \frac{3dx}{(x-1)(3+2\sqrt{x-1})} = \int \frac{3 \times 2u du}{(u^2+1-1)(3+2u)}$	M1	1.1b
	Finds correct limits e.g. $p = 2, q = 3$	B1	1.1b
	$= \int \frac{3 \times 2u du}{u^2(3+2u)} = \int \frac{6 du}{u(3+2u)} *$	A1*	2.1
		(4)	
(b)	$\frac{6}{u(3+2u)} = \frac{A}{u} + \frac{B}{3+2u} \Rightarrow A = \dots, B = \dots$	M1	1.1b
	Correct PF. $\frac{6}{u(3+2u)} = \frac{2}{u} - \frac{4}{3+2u}$	A1	1.1b
	$\int \frac{6 du}{u(3+2u)} = 2 \ln u - 2 \ln(3+2u) \quad (+c)$	dM1 A1ft	3.1a 1.1b
	Uses limits $u = "3", u = "2"$ with some correct $\ln$ work leading to $k \ln b$ E.g. $(2 \ln 3 - 2 \ln 9) - (2 \ln 2 - 2 \ln 7) = 2 \ln \frac{7}{6}$	M1	1.1b
	$\ln \frac{49}{36}$	A1	2.1
		(6)	
(10 marks)			
Notes: Mark (a) and (b) together as one complete question			

(a)

B1: $dx = 2u du$ or exact equivalent. E.g. $\frac{dx}{du} = 2u, \frac{du}{dx} = \frac{1}{2}(x-1)^{\frac{1}{2}}$

M1: Attempts a full substitution of $x = u^2 + 1$, including $dx \rightarrow \dots du$ to form an integrand in terms of u .

Condone slips but there should be an attempt to use the correct substitution on the denominator.

B1: Finds correct limits either states $p = 2, q = 3$ or sight of embedded values as limits to the integral

A1*: Clear reasoning including one fully correct intermediate line, including the integral signs, leading to the given expression ignoring limits. So B1, M1, B0, A1 is possible if the limits are incorrect, omitted or left as 5 and 10.

(b)

M1: Uses correct form of PF leading to values of A and B .

A1: Correct PF $\frac{6}{u(3+2u)} = \frac{2}{u} - \frac{4}{3+2u}$ (Not scored for just the correct values of A and B)

dM1: This is an overall problem solving mark. It is for using the correct PF form and integrating using $\ln$ s.

Look for $P \ln u + Q \ln(3+2u)$

A1ft: Correct integration for their $\frac{A}{u} + \frac{B}{3+2u} \rightarrow A \ln u + \frac{B}{2} \ln(3+2u)$ with or without modulus signs

M1: Uses their 2 and 3 as limits, with at least one correct application of the addition law or subtraction law leading to the form $k \ln b$ or $\ln a$. PF's must have been attempted. Condone bracketing slips. Alternatively changing the u 's back to x 's and use limits of 5 and 10.

A1: Proceeds to $\ln \frac{49}{36}$. Answers without working please send to review.

Q14.

Question	Scheme	Marks	AOs
(a)	Uses the model to state $\frac{dV}{dt} = -c$ (for positive constant c)	B1	3.1b
	Uses $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ with their $\frac{dV}{dt} = -c$ and $\frac{dV}{dr} = 4\pi r^2$	M1	2.1
	$-c = 4\pi r^2 \times \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = -\frac{c}{4\pi r^2} = -\frac{k}{r^2}$ *	A1*	2.2a
		(3)	
(b)	$\frac{dr}{dt} = -\frac{k}{r^2} \Rightarrow \int r^2 dr = \int -k dt$ and integrates with one side "correct"	M1	2.1
	$\frac{r^3}{3} = -kt (+\alpha)$	A1	1.1b
	Uses $t = 0, r = 40 \Rightarrow \alpha = \dots$ $\alpha = \frac{64000}{3}$	M1	1.1b
	Uses $t = 5, r = 20$ & $\alpha = \dots \Rightarrow k = \dots$	M1	3.4
	$r^3 = 64000 - 11200t$ or exact equivalent	A1	3.3
		(5)	
(c)	Uses the equation of their model and proceeds to a limiting value for t E.g. "64000 - 11200t" ... $0 \Rightarrow t \dots$	M1	3.4
	For times up to and including $\frac{40}{7}$ seconds	A1ft	3.5b
		(2)	
(10 marks)			
Notes:			

(a)

B1: Uses the model to state $\frac{dV}{dt} = -c$ (for positive constant c).

Any "letter" is acceptable here including k .

Note that $\frac{dV}{dt} = c$ is B0 unless they state that c is a negative constant.

M1: For an attempt to use $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ with their $\frac{dV}{dt}$ and $\frac{dV}{dr} = 4\pi r^2$

Allow for an attempt to use $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ with their $\frac{dV}{dt}$ and $\frac{dV}{dr} = \lambda r^2$ (Any constant is fine)

There is no requirement to use the correct formula for the volume of a sphere for this mark.

A1*: Proceeds to the given answer with an intermediate line equivalent to $\frac{dr}{dt} = -\frac{c}{4\pi r^2}$

If candidate started with $\frac{dV}{dt} = -k$ they must provide a minimal explanation how

$$\frac{dr}{dt} = -\frac{k}{4\pi r^2} \rightarrow \frac{dr}{dt} = -\frac{k}{r^2} \text{ . E.g. } \frac{1}{4\pi} \text{ is a constant so replace } \frac{k}{4\pi} \text{ with } k$$

It is not necessary to use the full formula for the volume of a sphere, eg allow $V = \kappa r^3$ but if it has been quoted it must be correct. So using $V = 4r^3$ can potentially score 2 of the 3 marks.

(b)

M1: For the key step of separating the variables correctly AND integrating one side with at least one index correct. The integral signs do not need to be seen.

A1: Correct integration E.g. $\frac{r^3}{3} = -kt(+\alpha)$ or equivalent. The $+\alpha$ is not required for this mark.

This may be awarded if k has been given a value.

M1: Uses the initial conditions to find a value for the constant of integration α

If a constant of integration is not present, or k has been given a pre defined value, then only the first two marks can be awarded in part (b)

The mark may be awarded if the equation has been adapted incorrectly. E.g. each term cube rooted.

M1: Uses the second set of conditions with their value of α to find k

This may be awarded if the equation has been adapted incorrectly. E.g. each term cube rooted.

A1: Obtains any correct equation for the model.

$$\text{E.g. } r^3 = 64000 - 11200t \text{ or exact equivalent such as } \frac{r^3}{3} = \frac{64000}{3} - \frac{11200}{3}t$$

ISW after sight of a correct answer. Condone recurring decimals e.g. $21333.\dot{3}$ for $\frac{64000}{3}$

Do not award if **only** the rounded/truncated decimal equivalents to say $\frac{64000}{3}$ is used.

(c)

M1: Recognises that the model is only valid when $r \geq 0$ and uses this to find t . Condone $r > 0$

Award for an attempt to find the value of t when $r = 0$. See scheme.

It must be from an equation of the form $ar^n = b - ct$, $a, b, c > 0$ which give +ve values of t .

A1ft: Allow valid for times up to (and including) $\frac{40}{7}$ seconds, 5.71 seconds. Allow $t < \frac{40}{7}$ or $t \leq \frac{40}{7}$

There is no requirement for the left hand side of the inequality, 0

States invalid for times greater than $\frac{40}{7}$ seconds, 5.71 seconds.

Follow through on their equation so allow $t < \text{their } \frac{64000}{11200}$ as long as this value is greater than 5

($t = 5$ is one of the values in the question)

Question	Scheme	Marks	AOs
(a)	$x^2 \tan y = 9 \Rightarrow 2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$ Q15.	M1 A1	3.1a 1.1b
	Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$	M1	1.1b
	$\frac{dy}{dx} = \frac{-2x \times \frac{9}{x^2}}{x^2 \left(1 + \frac{81}{x^4}\right)} = \frac{-18x}{x^4 + 81} \quad *$	A1*	2.1
		(4)	
(b)	$\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$ $\Rightarrow \frac{d^2y}{dx^2} = \frac{-18 \times (x^4 + 81) - (-18x)(4x^3)}{(x^4 + 81)^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2} \text{ o.e.}$	M1 A1	1.1b 1.1b
	States that when $x < \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} < 0$ when $x = \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} = 0$ AND when $x > \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} > 0$ giving a point of inflection when $x = \sqrt[4]{27}$	A1	2.4
		(3)	
(7 marks)			
Notes:			

(a)

M1: Attempts to differentiate $\tan y$ implicitly. Eg. $\tan y \rightarrow \sec^2 y \frac{dy}{dx}$ or $\cot y \rightarrow -\operatorname{cosec}^2 y \frac{dy}{dx}$

You may well see an attempt $\tan y = \frac{9}{x^2} \Rightarrow \sec^2 y \frac{dy}{dx} = \dots$

When a candidate writes $x^2 \tan y = 9 \Rightarrow x = 3 \tan^{\frac{1}{2}} y$ the mark is scored for $\tan^{\frac{1}{2}} y \rightarrow \dots \tan^{\frac{3}{2}} y \sec^2 y$

A1: Correct differentiation $2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$

Allow also $\sec^2 y \frac{dy}{dx} = -\frac{18}{x^3}$ or $2x = -9 \operatorname{cosec}^2 y \frac{dy}{dx}$ amongst others

M1: Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$

A1*: Proceeds correctly to the given answer of $\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$



(b)

M1: Attempts to differentiate the given expression using the product or quotient rule.

For example look for a correct attempt at $\frac{vu' - uv'}{v^2}$ with $u = -18x, v = x^4 + 81, u' = \pm 18, v' = \dots x^3$

If no method is seen or implied award for $\frac{\pm 18 \times (x^4 + 81) \pm 18x(ax^3)}{(x^4 + 81)^2}$

Using the product rule award for $\pm 18(x^4 + 81)^{-1} \pm 18x(x^4 + 81)^{-2} \times cx^3$

A1: Correct simplified $\frac{d^2y}{dx^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2}$ o.e. such as $\frac{d^2y}{dx^2} = \frac{54x^4 - 1458}{(x^4 + 81)^2}$

Alternatively score for showing that when a correct (unsimplified) $\frac{d^2y}{dx^2} = 0 \Rightarrow x^4 = 27 \Rightarrow x = \sqrt[4]{27}$

Or for substituting $x = \sqrt[4]{27}$ into an unsimplified but correct $\frac{d^2y}{dx^2}$ and showing that it is 0

A1: Correct explanation with a minimal conclusion and correct second derivative.

See scheme.

It can be also be argued from $x^4 < 27$, $x^4 = 27$ and $x^4 > 27$ provided the conclusion states that the point of inflection is at $x = \sqrt[4]{27}$

Alternatively substitutes values of x either side of $\sqrt[4]{27}$ and at $\sqrt[4]{27}$, into $\frac{d^2y}{dx^2}$, finds all three values and makes a minimal conclusion.

A different method involves finding $\frac{d^3y}{dx^3}$ and showing that $\frac{d^3y}{dx^3} \neq 0$ and $\frac{d^2y}{dx^2} = 0$ when $x = \sqrt[4]{27}$

FYI $\frac{d^3y}{dx^3} = \frac{23328x^3}{(x^4 + 81)^3} = 0.219$ when $x = \sqrt[4]{27}$

Alternative part (a) using arctan

M1: Sets $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times \dots$ where ... could be 1

A2: $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times -\frac{18}{x^3}$

A1*: $\frac{dy}{dx} = \frac{1}{1 + \frac{81}{x^4}} \times -\frac{18}{x^3} = \frac{-18x}{x^4 + 1}$ showing correct intermediate step and no errors.